

SIDE QUEST: THE LEANING TOWER OF REGOLITH

A XR-Based interactive construction challenge at the Australian Rover Challenge 2026.



1. What is the Side Quest?

The Side Quest is an interactive, short-duration challenge where participants supervise a robotic arm building a vertical tower of blocks. Rather than directly controlling the robot via joystick, participants operate in a supervisory role: they monitor the robot's approach, evaluate the alignment of the incoming block against the existing stack, and decide whether to "Fix" the alignment (incurring a time penalty) or "Drop" the block (risking structural instability).

The task is framed as a "Risk vs. Reward" stacking game. The robot intentionally introduces "drift" (random misalignment) during each approach. Participants must use XR overlays to judge if the drift is within the "Adaptive Zone" (Yellow) or if it constitutes a critical error (Red). Successful performance requires negotiating the trade-off between speed (accepting imperfect yellow placements) and stability (spending time to achieve green perfection).

2. The Challenge: "The Leaning Tower of Regolith"

At the core of the Side Quest is a vertical construction challenge driven by the geometric logic of the Rhombic Dodecahedron. Unlike standard cubic blocks, these faceted elements do not merely stack; they must "seat" into the geometry below. This creates a construction scenario where alignment is not binary (hit/miss) but continuous. A block can be placed imperfectly and still hold, but the error will propagate upward.

The challenge simulates the "Accumulated Drift" problem inherent in scaffoldless lunar masonry (Regolith). A minor rotational error at Block 1 may be negligible, but by Block 8, that same error shifts the center of gravity outside the base, leading to structural failure.

2.1 The Objective

The goal is to stack all 8 blocks as quickly as possible. The run ends only under three conditions:

1. Completion: All 8 blocks are successfully placed.
2. Physical Collapse: The tower actually falls over due to poor alignment choices.
3. Timeout: The participant exceeds the maximum session time (eg. 5 minutes).

2.2 The Risk vs. Reward

The robot intentionally introduces random "drift" (misalignment) during each approach. Using the XR overlays, the participant must judge the quality of the fit and decide whether to "Fix" (incurring a time cost) or "Drop" (accepting the risk). The decision is governed by three tectonic zones:

- **GREEN (Optimal):** The block is perfectly centered. Consequence: Maximum stability. Three consecutive greens, and the robot gains a speed boost for the next move.
- **YELLOW (Adaptive):** The block is off-center but within the geometry's "Adaptive Zone." Consequence: The tower holds, but the center of gravity shifts. No speed bonus.
- **RED (Critical):** The block is outside the safe seating threshold. Consequence: Structural Fault. The block is placed, but it introduces severe instability, significantly increasing the likelihood of a physical collapse on subsequent layers. The game continues, forcing them to manage a now-unstable structure.

This design shifts the cognitive task from simple error avoidance to strategic tolerance negotiation: players must decide how many "Red" or "Yellow" risks they can afford to take to top the leaderboard before physics takes over and the tower falls.

3. Multi-View Supervision

Participants supervise the task through two static camera views. Effective performance depends on switching views to check "Local Fit" versus "Global Stability."



Figure 1. (Left) Inspector view, (Right) Site Manager view.

Camera 1: "The Inspector" (High-Oblique View)

- Placement: Static, mounted high above the workspace looking down at a steep angle (60°-75°).
- Function: Joint Negotiation. This view provides a near-plan perspective to verify X/Y centering and rotation (Yaw) of the incoming block faces.
- XR Overlay: Displays the "Tolerance Ghost" (Green/Yellow/Red) showing the immediate fit of the joint.

Camera 2: "The Site Manager" (Side Elevation)

- Placement: Static, wide-angle camera mounted at table height (Side View).
- Function: Global Context. This view reveals the silhouette of the tower to show cumulative lean or "drift" that might be invisible from the top-down view.
- XR Overlay: Displays the "Center of Gravity" line. If the line deviates too far from the central axis, the tower is at risk.

Crucially: The robot arm may obscure one view during the approach, forcing the user to switch perspectives to verify the placement confidence.

4. How Participants Interact

4.1. Roles

- Participant: Site Supervisor. They do not steer the robot. They issue high-level commands: "Nudge," "Fix," or "Commit/Drop."
- Robot: Executes pick-and-place actions with injected "noise" (drift).
- Facilitator: Resets the physical blocks between runs.

4.2. Flow of a Run

1. Briefing: "Build the tower as fast as possible. If the Ghost is Red, the tower might fall. If it's Yellow, it's risky but fast. If it's Green, it's safe."
2. The Approach: The robot picks a block and hovers 2cm above the stack. It adds a random offset (e.g., 3mm Left).
3. Tolerance Negotiation:
 - The User sees a Yellow Ghost.
 - Choice A (Risk): User presses "DROP." The robot places it immediately. Time saved, but stability drops.
 - Choice B (Caution): User presses "FIX." The robot spends 3 seconds re-aligning to Green. Stability preserved, but time lost.
4. Completion: The run ends when 8 blocks are stacked, the tower collapses, or time runs out.

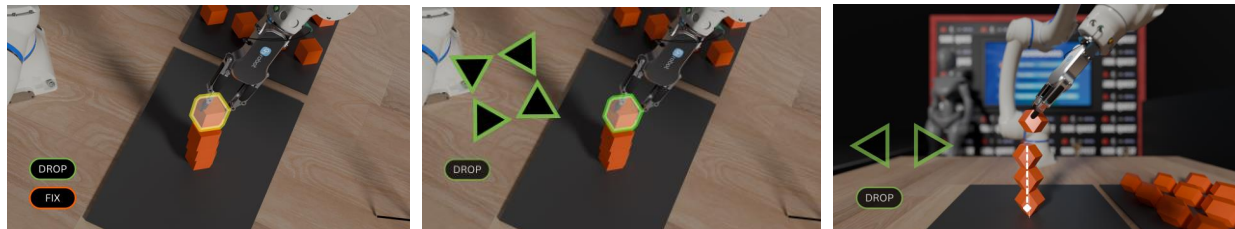


Figure 2. (Left) The approach, (Middle and Right) if participant chooses to fix, they can nudge the block until green, and then drop.

5. Robot Autonomy and Human Authority

The robot operates with Sliding Autonomy, picking up blocks from the pick-up plate and bringing it to the drop spot, while it relinquishes the final drop authority to the human.

- The Robot: Handles all sensing, inverse kinematics, gripping, and movement. It "wants" to place the block but is "clumsy" (simulated drift).
- The Human: Acts as the Gatekeeper. The robot cannot release the block until the human commits.
- The Innovation: Unlike teleoperation (steering), the human is modifying the State of the action (Safe vs. Risky) rather than the Path.

6. Functional Roles and System Architecture

6.1 Functional Roles

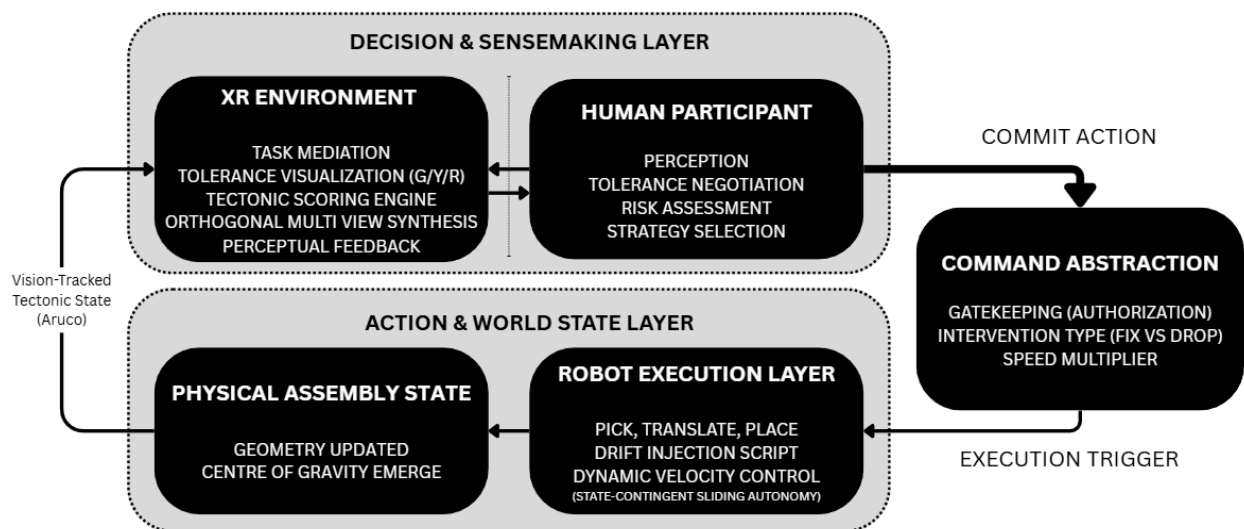


Figure 3. Functional Roles

The functional architecture of the Side Quest is structured to emphasize an asymmetric relationship between human cognition and robotic execution, ie robot proposes an action, human authorizes, and robot executes, as opposed to them sharing similar roles. The system is organized into three interdependent layers that separate high-level tectonic judgment from low-level physical actuation (Figure 3).

6.1.1 Decision and Sensemaking Layer (Human–XR Mediator)

The upper layer (figure x) represents the cognitive core of the construction system. In this layer, the XR Environment functions as an active communication mediator between the robot and the human, rather than a passive display. It ingests stereoscopic and wide-angle visual data from the orthogonal OAK-D camera array to calculate the real-time delta between the physical assembly state and the architectural target. This data is externalized through Graded Tolerance Fields (Green/Yellow/Red), which visualize the "Adaptive Zone". Additionally, the system monitors tower height and time, providing the necessary feedback loops for competition.

The Human Participant occupies the role of a Strategic Supervisor. Relieved of manual teleoperation, the participant's cognitive resources are redirected toward Tolerance Negotiation. They must synthesize the local alignment data (the ghost gradient) with the global stability context (the tower's center of gravity) to perform a continuous risk assessment. The human role is thus defined by the decision to either accept an imperfect but fast "Yellow" placement or intervene with a costly "Fix" command to restore structural equilibrium.

6.1.2 Command Abstraction (The Gatekeeper)

The Command Abstraction serves as the critical bridge and control gate between digital intent and physical action. It enforces a "Human-as-Gatekeeper" protocol where the robot is permitted to approach and hover, but is restricted from final commitment. This layer encodes the participant's choice into discrete, irreversible command packets: FIX (Corrective) or DROP (Commit), while simultaneously passing dynamic parameters, such as the Speed Multiplier earned through precision streaks, down to the robot control stack.

6.1.3 Action and World-State Layer (Robot-Physical Environment)

The lower layer represents the embodied reality of the lunar construction site. The Robot Execution Layer operates under State-Contingent Autonomy, managed by a Raspberry Pi 5. To create a meaningful supervisory challenge, this layer utilizes a Drift Injection Script that introduces randomized Gaussian noise into the robot's proposed coordinates, simulating the unpredictability of mobile rover platforms. A Dynamic Velocity Controller adjusts the manipulator's speed based on the coordination quality reported from the upper layers.

The Physical Assembly State is governed by the specific facets of the Rhombic Dodecahedron blocks. Unlike simple cubes, these geometries produce emergent constraints; a placement that is technically "successful" may still result in a Center of Gravity (COG) Drift that destabilizes the entire tower. This physical state is continuously re-ingested into the system via Vision-Track Tectonic State (Aruco), closing the perception-

action loop and ensuring that the XR environment remains a high-fidelity Digital Twin of the evolving construction.

6.2 System Architecture

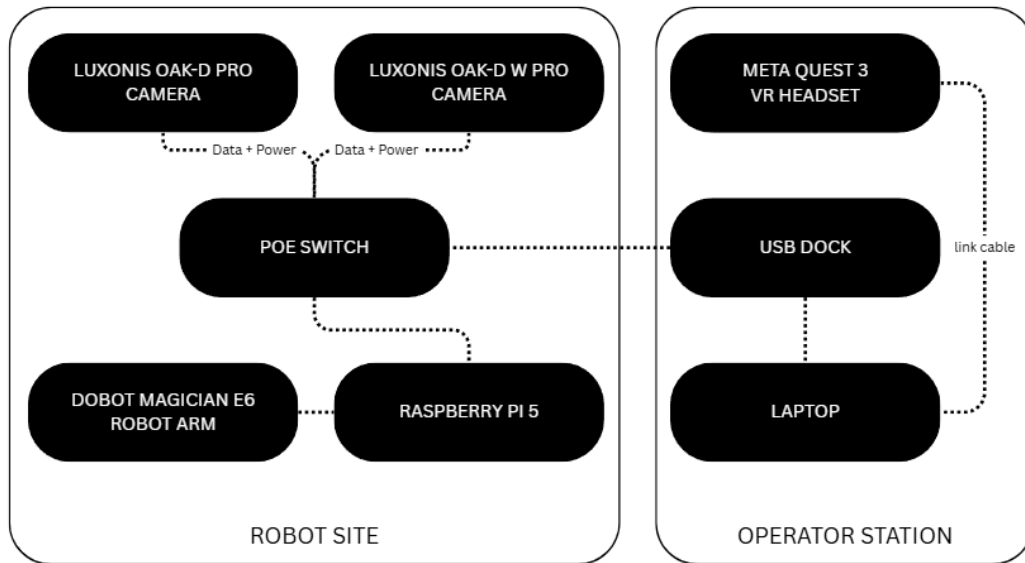


Figure 4. Interconnectivity map of hardware

To support the real-time processing demands of stereoscopic XR and semi-autonomous robotic control, the research utilizes a distributed hardware architecture (Figure 4). This interconnectivity map illustrates the physical networking and data protocols required to facilitate 'Alignment Tectonics' in a remote construction environment.

Central to this architecture is a High-Bandwidth Network Hub (PoE Switch), which serves as the primary data conduit. The two Luxonis OAK-D Pro cameras are integrated via Power-over-Ethernet (PoE), providing high-resolution stereoscopic and wide-angle vision feeds while simplifying cable management at the Roseworthy site. Physical execution is offloaded to a Raspberry Pi 5, which hosts the ROS 2 control stack and manages the Dobot Magician E6 manipulator. This distributed approach ensures that the robotic control loop remains isolated from the interface rendering, enhancing system stability and safety.

The Operator Station consists of a high-performance laptop running the Unity XR Runtime. This node ingests the dual-camera streams through the network hub to calculate the real-time tectonic alignment gradients. Finally, the Meta Quest 3 is connected via a dedicated Link cable to provide the low-latency, stereoscopic pass-through necessary for the supervisor to negotiate the architectural adaptive zones of the construction task.

The system is realized through a four-layer architecture designed for low-latency video transmission and reliable robotic control, as shown in figure 5.

Layer 1: Unity XR Runtime (The Decision Engine)

- Visual Pipeline: Ingests dual video streams from the OAK-D cameras. The "Inspector" feed (High-Oblique) and "Manager" feed (Side) are rendered as selectable viewports within the Quest 3 environment.
- Alignment Logic: Calculates the Euclidean distance and rotational delta between the Actual Base (Aruco tracking) and Proposed Target.
- Gradient Shader: Maps the calculated delta to the visual overlay color (eg. $\Delta < 2\text{mm}$ = Green; $\Delta < 5\text{mm}$ = Yellow).
- Game State: Manages the Leaderboard, Timer, Combo Multiplier (Speed Boost), and Structural Integrity Bar.

Layer 2: ROS Interface Middleware

- Acts as the communication bridge using `rosbridge_suite`.
- Transmits high-level commands from Unity.
- Receives robot state data (Joint angles, Gripper status) to update the Digital Twin representation.

Layer 3: Robot Control Stack (ROS Nodes)

- Drift Generator Node: The core logic for the challenge. For every new block, this node calculates the ideal stacking coordinate and adds a randomized Gaussian noise vector (x, y, yaw) to generate the "Proposed Pose."
- Motion Planner: Calculates the inverse kinematics for the Dobot E6 to reach the "Hover" point and the final "Drop" point.
- Safety Monitor: Ensures the robot does not exceed the Z-height limits or collide with the camera gantry.

Layer 4: Physical Assembly State

- Hardware: Dobot Magician E6 with custom friction-paddle grippers and 3D-printed Rhombic Dodecahedron blocks (approx. 40mm diameter).
- Physics: The physical interaction of the faceted blocks determines the actual stability. If a "Red" block is placed, the physics of the stack (center of gravity) changes, influencing the starting state for the next block.

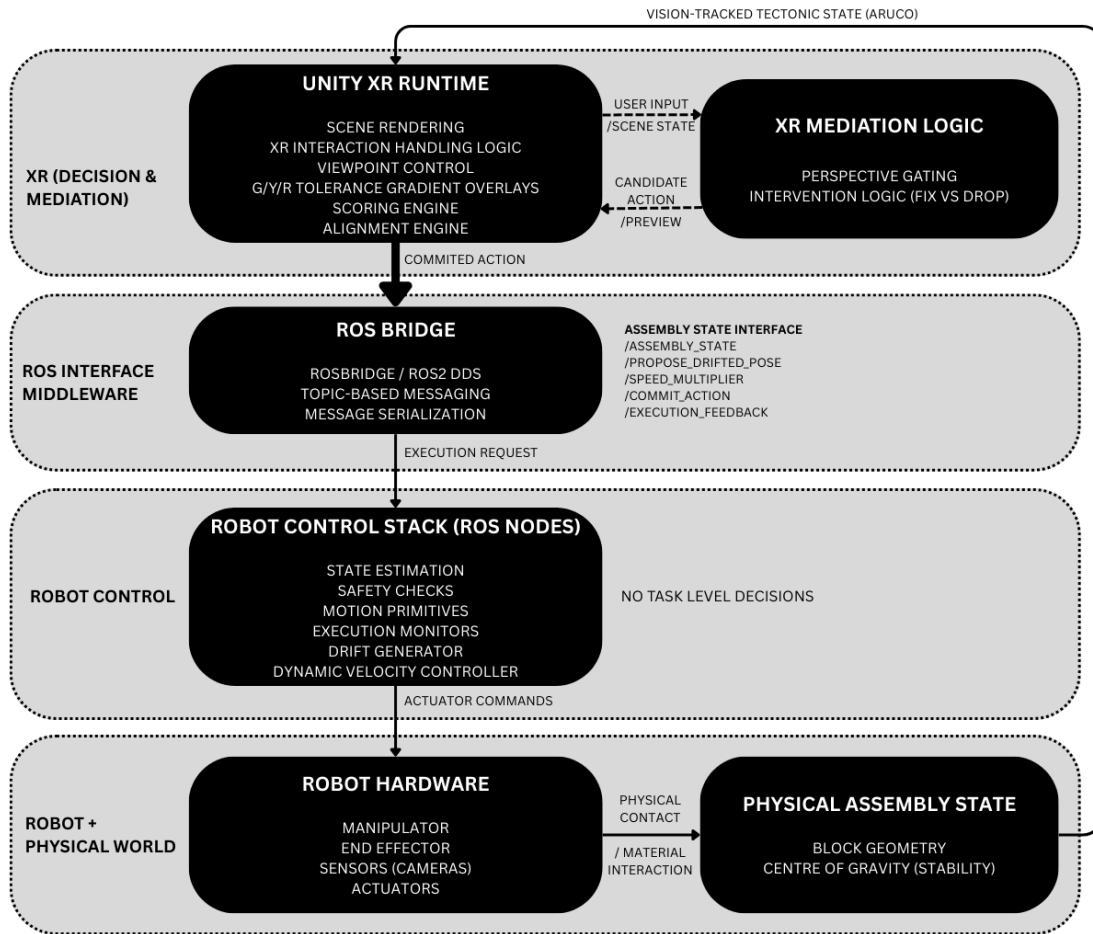


Figure 5. System Architecture for XR-Mediated Human-Robot Assembly

7. Scoring, Leaderboard, and Competition

7.1 The Ranking Logic

The leaderboard ranks participants based on a hierarchy of two metrics:

1. **Primary Metric: Tower Height (Max 8).** The absolute number of blocks successfully stacked and stable at the end of the run. Example: A participant who stacks 8 blocks (even slowly) always ranks higher than a participant who stacks 7 blocks quickly. This prioritizes structural success over speed.
2. **Tie-Breaker: Completion Time.** If two participants achieve the same height (e.g., both stack 8 blocks), the participant with the faster time ranks higher. Strategic Implication: This forces the "Risk vs. Reward" decision. To get the fastest time, players must decide when to accept "Yellow" (imperfect) placements to save time, without pushing the tower into "Red" (Collapse).

7.2 The Role of "Tectonic Zones" (Green/Yellow/Red)

While "Quality Points" are not explicitly added to the score, the Tectonic Zones function as the critical enablers of performance. They represent the "Health" of the tower.

- Green (Optimal): Game Effect - Speed Boost. Achieving 3 consecutive Green placements triggers "Combo Mode," increasing the robot's movement speed by 10-20%. This is how skilled players achieve the fastest times.
- Yellow (Adaptive): Game Effect - Survival. The placement is accepted, but the center of gravity shifts. The tower becomes harder to stack higher.
- Red (Critical): Game Effect - Physics Risk. The system allows the placement, but the geometric misalignment makes a physical collapse highly probable on the next block. If the tower falls, the Height Score drops to the level of the remaining stump / last successful placement.

7.3 The Leaderboard Display

The 65" public display features a simple, arcade-style list:

Rank	Pilot Name	Height	Time
1	Alice	8	01:45
2	Bob	8	02:10
3	Charlie	6	01:15
...	...	...	...

Figure 6. Leaderboard mock-up

8. Physical and Technical Setup

The system is deployed as a compact, self-contained demonstration booth at ARC 2026. The physical setup comprises a robotic arm mounted on a dedicated work-table, enclosed by a low-profile safety barrier that separates the active manipulation zone from participants while maintaining full spectator visibility.

8.1 Vision and Camera Configuration

Crucial to the "Leaning Tower" mechanic, the setup utilizes two static Luxonis OAK-D cameras arranged to provide orthogonal validation views, minimizing the risk of robot self-occlusion while maximizing alignment visibility:

1. **Camera 1: "The Inspector" (High-Angle Oblique):** Mounted on a gantry or overhead truss at a steep angle (**approx. 75°**). This view acts as the "Plan" perspective, allowing

the participant to judge the X/Y centering and rotational alignment of the faceted blocks.

2. **Camera 2: "The Site Manager" (Side Elevation):** Mounted on a tripod at table height. This view provides the "Elevation" perspective, critical for detecting the cumulative lean (drift) of the tower and the center of gravity relative to the base.

8.2 Robotics and End-Effectors

The Dobot Magician E6 is equipped with a custom-fabricated gripper. Unlike standard parallel jaws, these rubberized, flat-faced paddles are designed to "palm" the 40mm Rhombic Dodecahedron blocks, ensuring a secure grip on the faceted geometry while allowing the robot to introduce controlled "drift" (randomized offsets) without dropping the payload.

8.3 Environmental Simulation

The Side Quest booth utilizes Standard Overhead Lighting consistent with an indoor convention or laboratory setting. This controlled lighting environment, as opposed to lunar lighting simulation, is a deliberate choice for the Side Quest to ensure the absolute reliability of the computer vision tracking (Aruco markers) during continuous public operation.

8.4 Layout of the booth

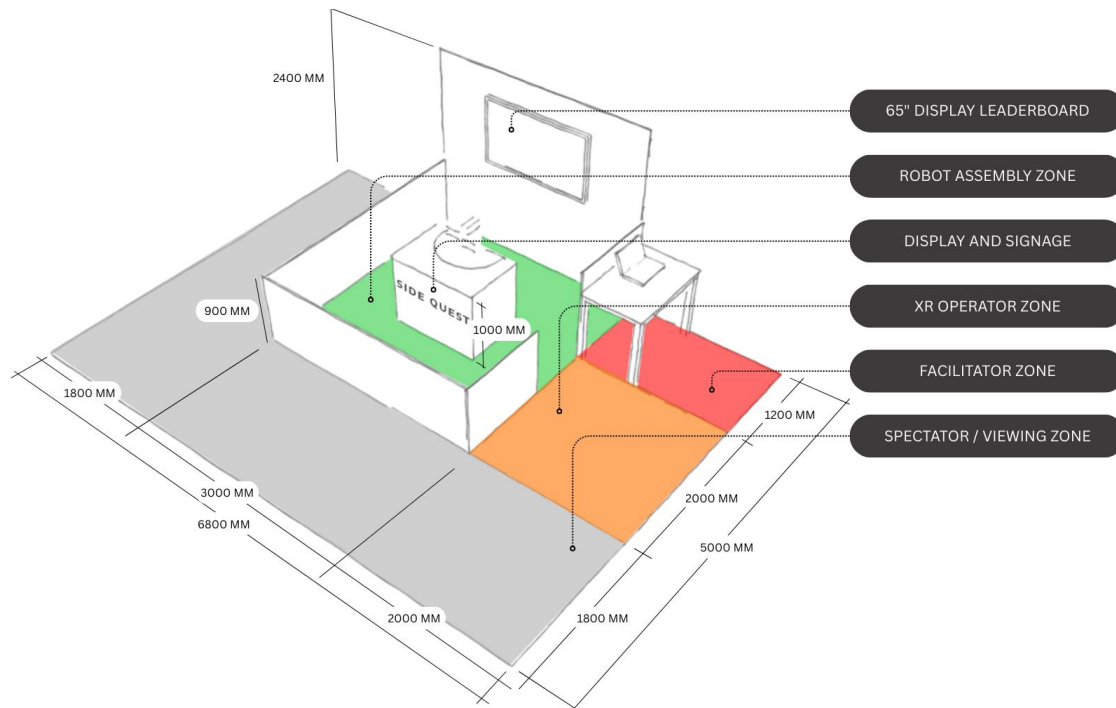


Figure 7. Layout of the Side quest booth.

Participants stand at a "XR operator zone" located outside the safety barrier (Figure 7). The zone is adjacent to the facilitator zone, which houses the VR PC and cable management for the Meta Quest 3 headset. In the robot assembly zone, a large 65" Display mounted behind the robot mirrors the participant's XR view for spectators and displays the live Leaderboard, creating a focal point for competition.

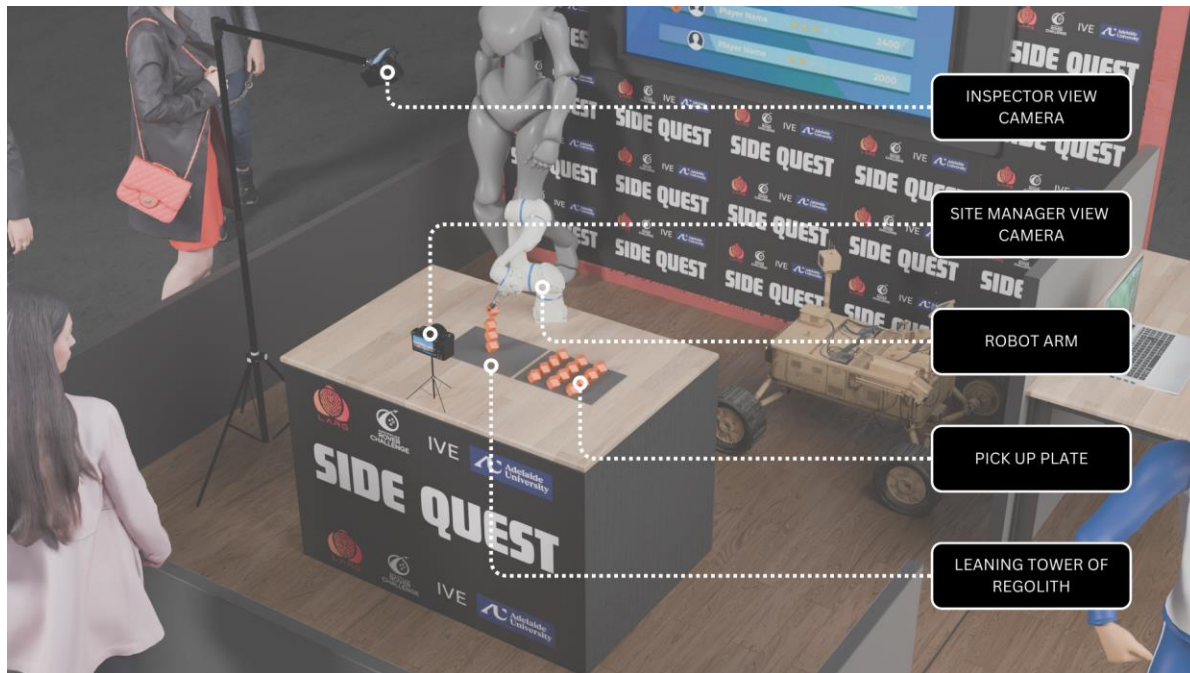


Figure 8. Set-up of the robot working table.

9. Design Scope and Ambition

The table below summarizes three internally consistent configurations for the Phase 2A Side Quest, providing a flexible framework that scales from a minimal validation state to an aspirational future vision.

The MVP configuration represents the baseline entry point, focusing on a 3-4 block vertical stack under standard overhead lighting with binary "Pass/Fail" tectonic seating logic. In this tier, the XR interface provides a monoscopic video feed in the VR headset with a static wireframe target and simple UI beeps for feedback. The human-robot collaboration is characterized by a single construction bot operated via facilitator mediation (Wizard-of-Oz), with the human supervisor utilizing a basic gesture controller to monitor the static alignment task.

The Middle Ground configuration serves as the target for the Australian Rover Challenge, scaling the task to an 8–10 block tower where the supervisor must negotiate accumulated

drift and earn speed combos. This tier introduces a more sophisticated tectonic logic using a 3-zone gradient (Green/Yellow/Red) and a real-time tolerance indicator. The XR mode is upgraded to stereoscopic video with egocentric spatial anchoring and an orthogonal view configuration consisting of a high 75° and a side-elevation camera. Audio feedback is enhanced with a satisfying "thud" on block release and a red alert siren for critical errors, while the robot operates under state-contingent sliding autonomy.

The Blue Sky configuration outlines the aspirational limit of the research, transitioning the task to a 15+ block multi-segment vault under high-contrast, low-angle lunar lighting and dynamic environmental physics. This tier utilizes a full Digital Twin XR environment with eye tracking for intent inference and a dynamic, real-time tolerance ghost. The tectonic logic is expanded to include a structural stress heatmap with adaptive audio cues, such as creaks and groans, to communicate instability. The HRC architecture becomes heterogeneous, involving a team of construction, dynamic observer, and transporter bots managed through multimodal voice and gesture commands and AI-suggested attention cues.

Table 1. Three-tier conceptualization of Side quest, ranging from a minimum viable state to a blue-sky state.

	Dimension	MVP	Middle Ground	Blue Sky
ARCH	Task	3-4 block vertical stack	8-10 block vertical tower	15+ block multi-segment vault
	Site fidelity	Side Quest Booth	Side Quest Booth	Side Quest Booth
	Lighting	Standard Overhead	Standard overhead	Low-angle High-contrast Lunar
	Tectonic Logic	Binary "Pass/Fail" seating	3-Zone Gradient (G/Y/R)	+ Stress Heatmap
	Coordination Stressor	Static target alignment	Accumulated Drift + Speed Combo	Dynamic environmental physics
	Rotation	No	No	Full 6-DOF
XR	XR Mode	Monoscopic Video in VR	Stereoscopic Video in VR	+ Digital Twin
	Eye Tracking	No	No	Yes (intent inference)
	XR Guidance	Static Wireframe Target	Static, real-time tolerance indicator	Dynamic, Real-time Tolerance Ghost
	Audio effects	Simple UI beeps	"Thud" on drop + Red Alert siren	+ Adaptive stress creaks/groans
	Spatial anchoring	None	Egocentric	+ Exocentric
	View Configuration	One fixed camera view	High 75° + Side	Multi-camera AI synthesis
	View Control	Manual switching	Manual switching	+ AI-suggested attention cues (no automatic switching)
HRC	Robot Count and Roles	1 construction bot	1 construction bot	1 construction bot + 1 dynamic observer bot+ other side bots like a transporter bot
	Robot Autonomy	operated by facilitator (WOz)	State-Contingent (Sliding Autonomy)	State-Contingent (Sliding Autonomy)

	HR Comm. mode	Gesture (controller)	Gesture (controller)	+ Voice
	Human Role	Supervisor	Supervisor	Supervisor
	Decision Authority	Robot. Human can intervene	Robot. Human can intervene	Robot. Human can intervene
	Perturbation	Scripted	Scripted	Scripted
	Safety gating	Facilitator mediated	Scripted	Scripted

10. Development Plan

This section outlines the strategic development plan required to transition the research from a controlled laboratory setting to the public-facing operational environment of the Australian Rover Challenge (ARC) 2026. This initiative is a collaborative effort between the Andy Thomas Centre for Space Resources (ATCSR) and the Australian Research Centre for Interactive and Virtual Environments (IVE) (Figure 9).



Figure 9. Example of visualization of real time data for mineral exploration produced by IVE for MinEx CRC.

10.1. XR Development Strategy

The XR development focuses on the transition from a visualization-only interface to a constitutive cognitive mediator. The strategy is structured into five iterative phases:

1. Stereoscopic Integration: Ingesting and calibrating dual OAK-D video streams into Unity to enable true-per-eye stereo perception within the Meta Quest 3, addressing the depth-perception gap identified in the SLR.

2. Spatial Anchoring: Establishing exocentric registration using Aruco markers to ensure XR overlays remain spatially consistent with the physical construction tower.
3. The Tolerance Engine: Developing a real-time shader-based visualization that calculates the delta between the robot's proposed pose and the architectural target, rendering the Green/Yellow/Red "Adaptive Zone" indicator. (This might also be done through the ROS generated drift).
4. Multi-View Orchestration: Implementing the orthogonal view-switching logic, allowing the user to toggle between the high-oblique "Inspector" camera and the side-elevation "Site Manager" camera.
5. Audio & UX Feedback: Integrating the "Thud" physics feedback for block release and the "Red Alert" siren to communicate structural risk via the auditory channel.

10.2 Robotics Development Strategy

The robotics stream focuses on building a predictable yet "clumsy" execution agent that requires human oversight. Development is conducted on the Dobot Magician E6 platform using ROS:

1. Deterministic Motion Primitives: Defining precise pick-and-place trajectories using the custom gripper to handle the 37.5 mm Rhombic Dodecahedron blocks.
2. Drift Injection Node: Developing a randomized perturbation script that forces the robot to arrive at slightly offset coordinates for every block, creating the necessary ambiguity for the human supervisor to resolve.
3. Sliding Autonomy Logic: Enable the robot to autonomously pick up the blocks and move them to the place spot and await human command. Worst case, scenario, we can use pre-programmed movements instead.
4. Combo boost: Implementing the "Speed Combo" multiplier, allowing the robot to increase execution velocity by 10-20% following consecutive "Green" placements, thereby gamifying high-trust coordination.
5. State Reporting: Ensuring the robot provides real-time symbolic feedback on its joint states and gripper status back to the Unity runtime for synchronization.
6. Scripted Safety Gating: Finalizing the automated safety routines and system resets to handle booth interruptions or physical collapses without damaging hardware.

10.3 System Implementation Plan

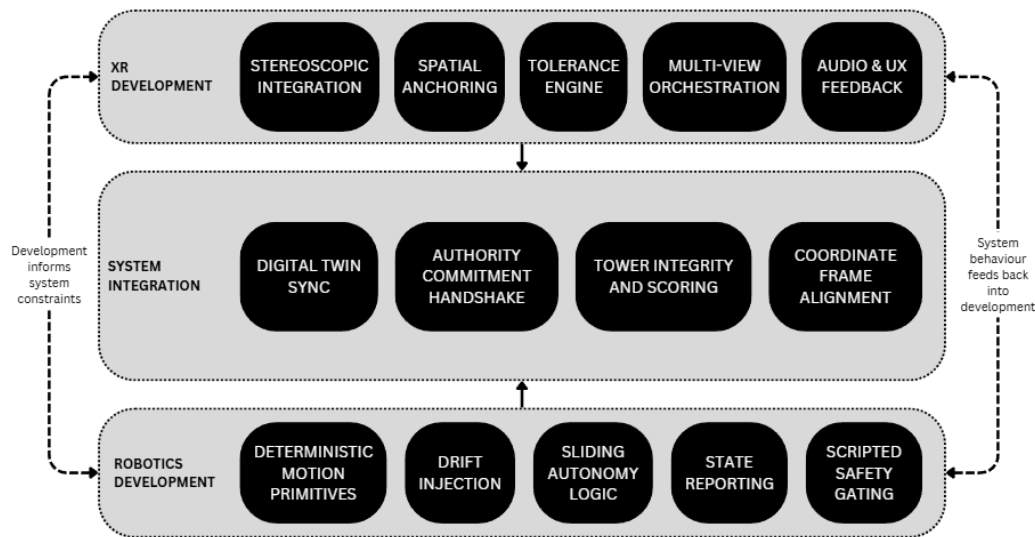


Figure 10. System Implementation plan.

The system architecture diagram (Figure 10) illustrates the multi-layered technical framework required to operationalize the Side Quest and Phase 2B experiments. The architecture is organized into three distinct but interdependent development streams: the XR Development layer (Human/Perception), the Robotics Development layer (Physical/Execution), and the System Integration layer (Logic/Bridge). This hierarchy enforces the research's core principle of supervisory control, where the human-facing interface is architecturally separated from physical robotic actuation.

The System Integration layer serves as the central "brain" of the experiment, facilitating the bidirectional flow of data between the digital and physical worlds. Within this layer, Digital Twin Sync ensures the virtual representation in the Meta Quest 3 headset remains latched to the physical state reported by the Luxonis cameras. The Authority Commitment Handshake acts as a critical gatekeeper, ensuring that no robotic action occurs without an explicit human command, thereby maintaining the asymmetry of decision authority. Tower Integrity and Scoring manages the real-time physics simulation and game logic, while Coordinate Frame Alignment maps the diverse spatial data from the cameras into the robot's localized XYZ coordinate space.

The system is characterized by iterative feedback loops, indicated by the dashed lines on either side of the diagram. The left loop, "Development informs system constraints," represents how the physical limits of the robotics (e.g., reach and drift) dictate the design of the XR tolerance engine and guidance cues. Conversely, the right loop, "System behavior feeds back into development," illustrates how the observed performance during assembly,

such as tracking loss under harsh lighting or structural collapses. It directly informs subsequent software refinements and architectural tectonic rules. This integrated approach ensures that the final "Alignment Tectonics" framework is both theoretically robust and operationally resilient.

10.4. Timeline

Table 2. Timeline for development

W	Dates	XR Development (Unity / XR Mediation)	Robotics Development (ROS / Robot)	Booth / Integration
1	Feb 10 – Feb 14	COMPLETE: Unity Project setup; Stereo stream calibration		
2	Feb 17 – Feb 21	Lunar Control Room set-up; View-switching UI; Basic Wireframe overlays	Gripper 3D Printing; Motion Primitives	Finalize Booth layouts
3	Feb 24 – Feb 28	Tolerance Engine logic (G/Y/R); Marker tracking	Drift Injection Scripting; State reporting	Fabricate 15x Dodecahedrons
4	Mar 3 – Mar 7	Spatial Anchoring lock; Audio feedback integration	Sliding Autonomy; Combo Boost logic	Full System Integration
5	Mar 10 – Mar 14	Middle Ground Feature Freeze	Speed/Precision stress testing	Physical Booth Build out
6	Mar 17 – Mar 21	Performance optimization; UI polish	Safety gating validation; System Resets	Volunteer Briefing
7	Mar 23 – Mar 25	ARC 2026 Final Readiness	Final Hardware Soak Testing	Logistics to Site

10.5. Funding allocation and usage

Table 3. Funding allocation and usage

Category	Expense	Item / Description	Cost (AUD)	Funding Source
XR Development	XR software development	Researcher time for XR mediation logic, interaction design, and Unity development	\$5,000	In-kind: IVE
	XR hardware	Meta Quest 3 (PC-VR via Link)	\$700	Funded by AU
	Vision hardware	Luxonis OAK-D Pro stereo cameras (×2)	\$1,800	Funded by AU
	XR compute & peripherals	VR cables, mounts, calibration tools	0	Existing / in-kind
Robotics Development	Robotic arm	Dobot Magician E6	\$10,000	Funded by AU
	Robot control development	Researcher time for ROS integration, motion primitives, execution logic	\$5,000	In-kind: ATCSR
	End-effector & fixtures	Gripper mounts, alignment jigs, tolerancing aids	0	Existing / minor purchase
Booth & Physical Setup	Booth construction	Side Quest station structure (tables, framing, finishes)	TBD	ARC
	Display & signage	Leaderboard monitor + printed signage (SIDE QUEST, instructions, logos)	TBD	ARC
	Safety & accessibility	Barriers, floor markings, safety signage	TBD	ARC
	Furniture & infrastructure	Tables (robot + PC), chairs (×2), cable management	TBD	ARC
Total				